Pluralsight

Introduction to Machine Learning

Prepared for Intuit

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Machine Learning (ML) is the killer app for Big Data. The increased availability of compute resources through myriad cloud providers such as AWS and the proliferation of tools — from software libraries to ML "as a service" providers — have brought the power of ML to all kinds of programmers. However, to use ML effectively, one needs to understand the models, how to use them, how to evaluate their performance, and how to position ML models into a larger software ecosystem.

This course is intended for data scientists and software engineers. It maintains an optimal balance of theory and practice. For each machine learning concept, we first discuss the foundations, its applicability and limitations. Then we explain the implementation and use, and specific use cases. This is achieved through a combination of about 50% lecture and 50% lab work.

Duration

3 Days

Prerequisites

- Background in high school level statistics and math
- B.S. Computer Science degree or equivalent professional experience
- Basic Python experience (functions, classes, packages)
- Basic understanding of cloud computing and core cloud services
- Familiarity with programming in at least one language
- Be able to navigate Linux command line

Learning objectives

After this course, participants will be able to:

- Understand the complete ML lifecycle (data → modeling → deployment → monitoring)
- Build, evaluate, and optimize classical ML models
- Apply feature engineering and cross-validation correctly
- Perform hyperparameter tuning
- Interpret models using explainability techniques
- Deploy a trained ML model
- Understand MLOps fundamentals

This course includes:

- Introduction to machine learning principle
- Supervised learning algorithms focus, with minor coverage of unsupervised learning.

- Linear regression, Logistic regression, Random Forests, Gradient Boosted Decision Trees, Neural Net basics
- Introduction to data processing using NumPy and Pandas
- Introduction to data visualization using Matplotlib
- Intermediate hands-on experience with Scikit Learn
- Hands-On Labs and Exercise

This course does not include:
- Amazon SageMaker
- Advanced AWS
- Deep learning
- Proofs or theoretic motivation

Outline

1. Introduction to Modern Machine Learning
- Introduction to ML
- ML vs Deep Learning vs Generative AI
- The ML Product Lifecycle
- Offline vs Online learning
- Batch vs Real-time inference
- Real-world architecture overview

2. Data Collection and Preparation
- ETL vs ELT
- Structured vs unstructured data
- Data quality issues
- Bias in datasets
- Train/Validation/Test strategy

Hands-On:
- Loading dataset using Pandas / Polars
- Handling missing values
- Encoding categorical features
- Feature scaling

3. Exploratory Data Analysis (EDA)
- Correlation analysis
- Distribution inspection
- Detecting skewness
- Outliers
  Feature relationships
- Tools:
  Pandas
  Seaborn / Matplotlib

Lab:
- EDA on real-world business dataset

4. Supervised Learning – Regression & Classification
- Linear Regression

- Logistic Regression
- Regularization (L1/L2)
- Overfitting vs Underfitting
- Bias-Variance Tradeoff

Lab:

- Build regression model
- Build classification model
- Evaluate with appropriate metrics

5. Decision Trees and Ensemble Methods

- Decision Trees
- Random Forest
- Feature importance

Lab:

- Compare against Logistic Regression

6. Model Evaluation and Validation

- Accuracy vs Precision/Recall
- F1 Score
- Cross-validation (K-Fold)
- Stratified splits
- Confusion Matrix

Lab:

- Model comparison

7. Hyperparameter Tuning

- Grid Search
- Random Search
- When tuning matters

Lab:

- Compare tuned vs untuned models

8. Unsupervised Learning

- K-Means clustering
- Choosing K
- Dimensionality reduction (PCA overview)

Lab:

- Customer segmentation example